

## MATLAB SESSION -1



1. If  $u=2xyz$ ,  $v=3y^2$ ,  $w=x+z$  Evaluate 'J'

**Output:**  $j = [2*y*z, 2*x*z, 2*x*y]$   
 $[0, 6*y, 0]$   
 $[1, 0, 1]$   
 $\det = 12*y^2*z - 12*x*y^2$

2. Evaluate  $\int_0^6 \int_0^{x^3} 2x(x^2 + y^2) dy dx$ .

**Output:**  $q = 2.2068e+07$  or  $q = 22068000$

3. Evaluate  $\int_0^{19} \int_0^{18} \int_0^{17} (2x^2 + 3y^2 + 4z^2) dx dy dz$

**Output:**  $q = 5.5233e+06$  or  $q = 5523300$